

**PRACTICUM-III Plan**

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Batch:	2023-2027	Semester:	3rd sem
Branch:	ECE	Mentor Name:	Vikram Kumar

Embedded Systems and Robotics:

An Arduino-based automated lawn maintenance system that mows grass and detects obstacles using sensors, ensuring efficient, safe operation without manual intervention.

Automatic Grass Cutting Robot Using Arduino:

The project centers on designing an innovative robotic lawn mower that leverages Arduino technology to automate grass cutting and obstacle detection. Equipped with sensors, the mower can autonomously navigate a lawn, identifying and avoiding obstacles like trees, rocks, and garden furniture. This eliminates the need for manual lawn mowing, significantly reducing physical effort and saving time. The system's automation not only enhances efficiency but also ensures consistent lawn maintenance, making it ideal for both residential and commercial applications. Additionally, the project promotes sustainability by optimizing energy use and reducing the environmental impact of traditional lawn care methods, offering a modern, intelligent solution for hassle-free lawn upkeep.

Components

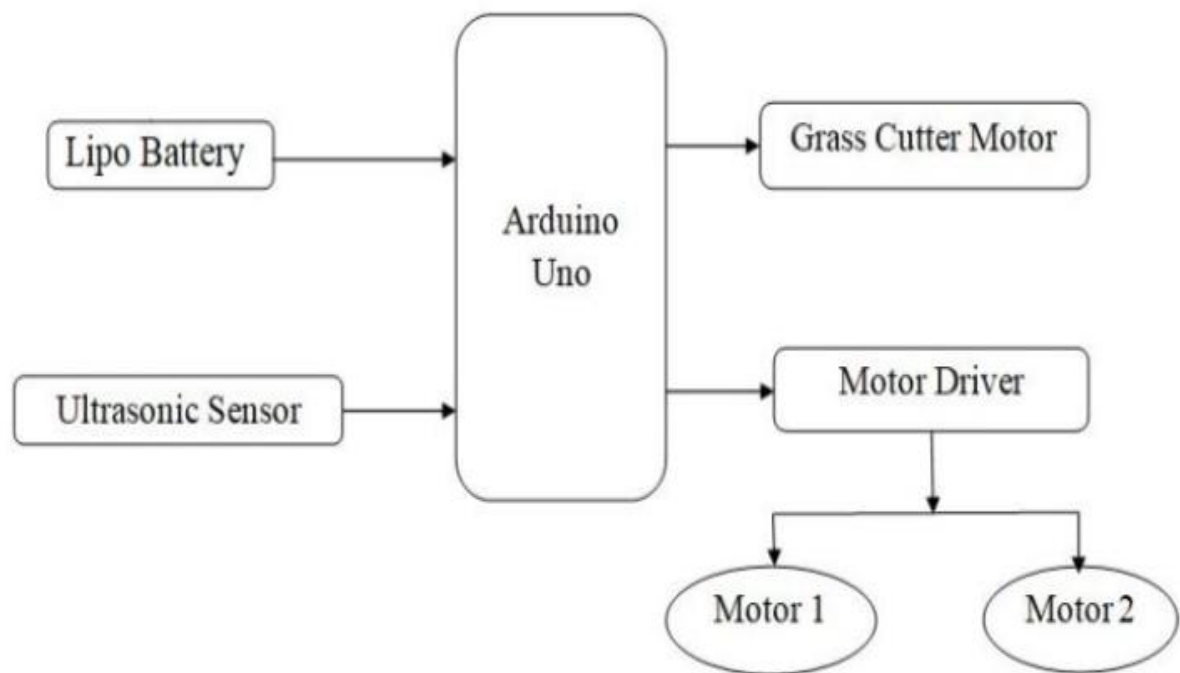
Here are the components needed for the Grass Cutting Robot (Lawn Mower):

1. Arduino UNO Board - 1
2. L293D Motor Driver Shield - 1
3. Ultrasonic Sensor HC-SR04 - 1
4. Ultrasonic Sensor Case/Holder - 1
5. DC Gear Motors- 4
6. BLDC Motor 100KV - 1

- 7. Servo Motor SG-90 - 1
- 8. ESC Module - 1
- 9. Servo Motor Tester - 1
- 10. 3-Pin Slide Switch - 1
- 11. X Type Cross Holder - 1
- 12. Robot Chassis - 1
- 13. 11.1V LiPo Battery - 1

Proposed Methodology:

The block diagram of Arduino Automatic Grass Cutting Robot shows how the different components are connected and how they interact with one another to perform the task of cutting grass automatically.



The Arduino board acts as the brain of the robot, controlling all the other components and executing the programmed instructions. The ultrasonic sensor is used to detect the location of grass and obstacles in the robot's path.

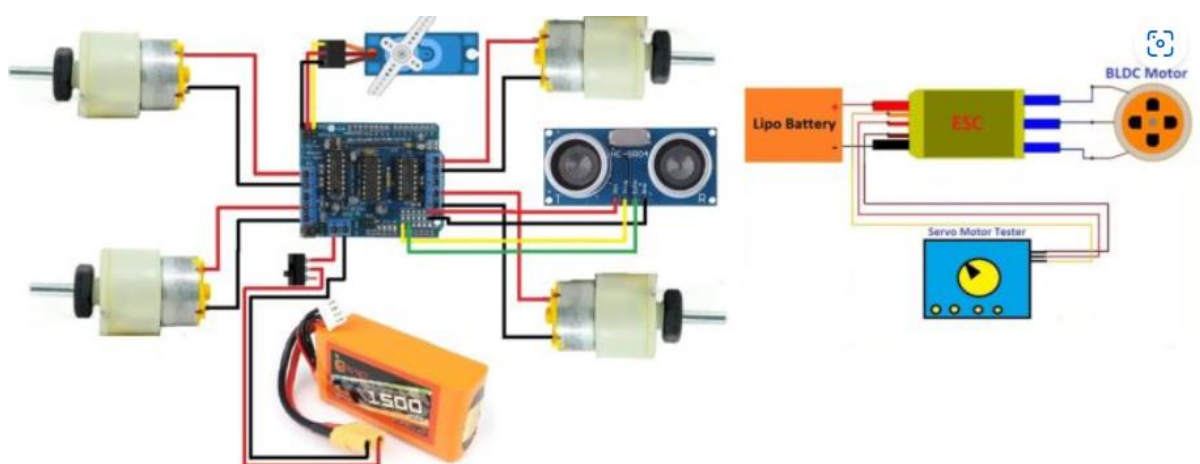
The L293D motor driver Shield is used to control the movement of the robot and is connected to the microcontroller. The 4 DC Gear motors are used to power the movement of the robot. And Servo Motor SG90 is attached at the head so pan motion to look for an obstacle on the left, right, and front.



For the cutting mechanism, a 100KV BLDC motor is used which rotates at a very high speed. This motor is connected to a cutting blade, which is used to cut the grass. The on/off mechanism and speed of the motor is controlled using the Servo Motor tester. You can use [BLDC Motor Controller](#) for this application. The robot requires a power source to operate, which can be a rechargeable battery or a power cord. In our case, we are using an 11.1V Lipo Battery.

Circuit Diagram & Connections

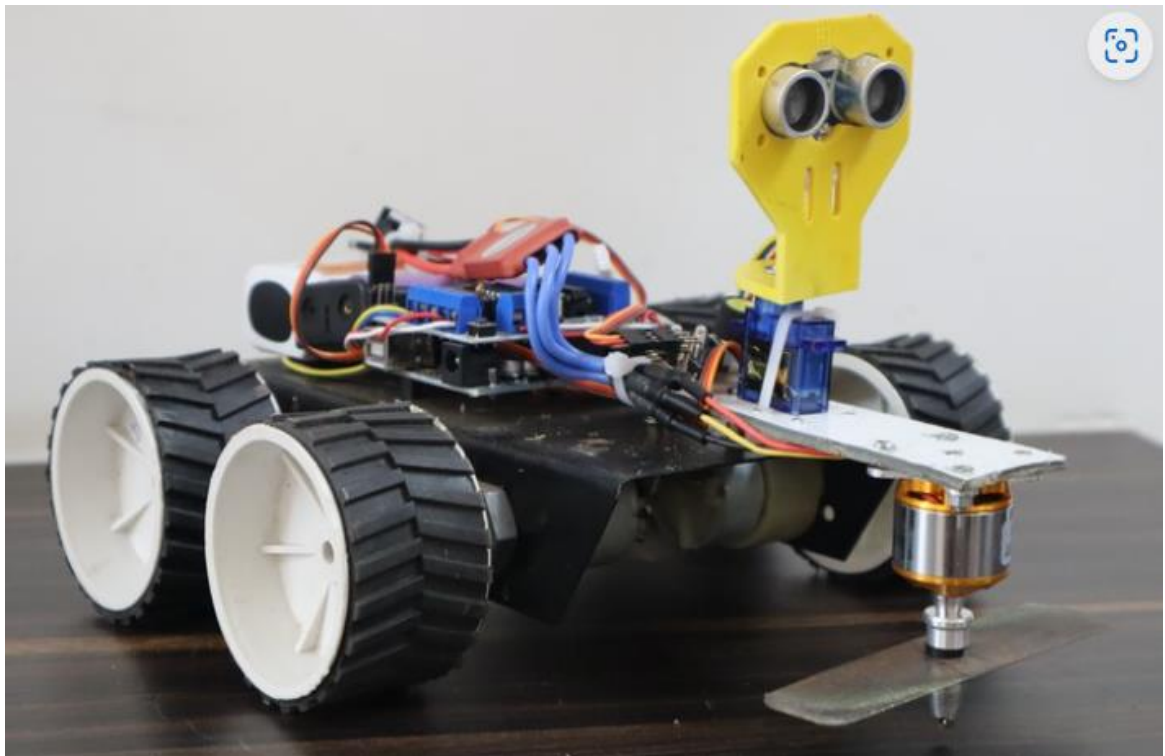
A circuit diagram of an automatic grass cutting robot using Arduino would show the connections between the different components of the robot. Here's an example of what the circuit diagram used for this project.



The L293D Motor Driver Shield is connected to the top of the Arduino UNO Board. The 4 DC motors are connected to the Motor Driver shield which is controlled via digital pins D1, D2, D3, and D4 of the Arduino UNO Board. The

Servo Motor is controlled via PWM Pin D10 of Arduino. The Ultrasonic Sensor HC-SR04 has TRIGN & ECHO Pins which are connected to A0 & A1 of the Arduino Board.

To control the BLDC Motor an ESC Module [Servo Tester](#) is used. The entire circuit is powered by an 11.1V Lithium-Ion Battery. The Arduino has a 5V regulator which converts the 11.1V to 5V Supply. The BLDC Motor directly operates at 11.1V.



Skillset Required to Solve/Address the Problem:

- **Technical Skills:**
 - Arduino programming
 - Circuit design and assembly
 - Sensor integration and calibration
 - Motor control and power management
 - Embedded systems troubleshooting

Week	Major Activities to be Completed
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	
Week 9	
Week 10	
Week 11	
Week 12	
Week 13	
Week 14	

Student Signature

Mentor Signature